RE IMAGINING CITIZEN SCIENCE MAKING ENVIRONMENTAL DECISIONS UNDER DURESS; CASE OF DZALEKA REFUGEE CAMP IN MALAWI

Grace Chilongo[1], Stanley Chilunga Chirwa[2], Brian Simbeye[3]

Lecturer, Share world open university. gracechilongo12@gmail.com[1]

Action for Social and Environmental Development. stanleychirwa7@gmail.com[2]

Action for Social and Environmental Development. briansimbeye9@gmail.com[3]

a. Project summary

i. Project participants

- The project involved refugees from Dzaleka, college students, lecturers, the government of Malawi and authors.

ii. Activities supported

- The project supported a mappathon event for the refugees.

iii. Project contribution to radical geographic scholarship and practice (outcomes and legacies)

- The project has revolutionized how the refugees analyze environmental conditions within the camp, thus from relying on third party information to being able to produce data on environmental conditions themselves, making sense out of it and making the necessary adjustments to adapt or mitigate the problem.
- The project has challenged the Dzaleka camp inhabitants to use locally available resources in researching more about their physical environment to support their decisions in environmental management.
- The project has made both the authorities and the camp inhabitants aware of the extent of environmental conditions within the camp and the need for a large-scale awareness and empowerment of the camp inhabitants on the development of environmental data and critical analytical skills.

iv. Major achievements

- The project managed to train 25 refugees and 10 college students on environmental data collection and analysis.
- Produced an environmental hazard map for the camp using participatory GIS.

v. Challenges

- Unavailability of apparatus for data collection on the local market
- High inflation which highly affected the budget
- Delays in releasing funds by the host institution.
- Bureaucracy from the government side led to delays in approvals for planned activities at the camp which delayed the project.

Major Achievements:

Empowered Marginalized Voices: The project successfully united university undergraduates and Dzaleka Refugee camp residents, creating a collaborative platform to co-design an environmental monitoring program.

Effective Implementation of Environmental Monitoring Program: Over a comprehensive five-month period, spanning both dry and wet seasons, the project seamlessly executed a citizen science initiative targeting water supplies, drainage, soil and air quality within the camp.

Creation of Multilingual Environmental Risk Map: A tangible and impactful outcome of the project is the development of an 'environmental risk' map, thoughtfully annotated in multiple languages: Swahili, Chichewa and English. This invaluable resource serves as a powerful tool for disseminating information to camp communities, the UN Refugee Agency, and the Malawi Department of Refugees.

Establishment of a Lasting Legacy: Community Research Lab: Capitalizing on the triumph of the camp project, the establishment of a 'Community Research Lab' at ShareWORLD University marks a lasting legacy. This lab, complemented by lesson plans in Chichewa and English, serves as the nucleus for an enduring critical geography initiative, ensuring continuous engagement in community-based research.

Pioneering the Expansion of Research Study Area: Building on the initial success within the camp, the project pioneers the expansion of research beyond the confines of the camp to nearby villages. The initiative sustains itself through annual classes of students, systematically broadening the scope of critical geographic inquiry and actively addressing ongoing challenges faced by marginalized communities.